



First Semester 2026-2027

Course Handout Part-II

Date: Aug 04, 2026

In addition to Part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : CS G514 / SS G514
Course Title : Object Oriented Analysis and Design
Instructor-in-Charge : MUKESH KUMAR ROHIL
Email : rohil@pilani.bits-pilani.ac.in **Chamber:** 6120-H
Course web page : Course page on Nalanda, the on-campus learning management system

Scope and Objective of the Course:

The course covers the object oriented analysis and design methodology, requirements-analysis using a use case model, use interaction models to capture the requirements and the design of the solution, in sufficient detail. The course will also cover topics like design patterns, implementation, testing, and architecture with respect to object oriented systems. After completion of the course the students will have good understanding of object oriented analysis and design using the unified process and will be able to apply the methodology to typical cases. The course aims at imparting a good understanding of the main UML models used in the use case driven unified process and their application to given problems.

Textbook:

T1. Object Oriented Modeling and Design using UML, Second Edition, Michael Blaha and James Rumbaugh, Pearson Education, Second Edition, 2007

Reference Books:

- R1.** Applying UML and Patterns, Third Edition, Craig Larman, Pearson Education, 2002
R2. Object Oriented Analysis and Design with Applications, Grady Booch, Addison-Wesley Second Edition, 1994
R3. Object Oriented Systems Development using Unified Modeling Language, Ali Behrami, McGraw Hill International Edition, 1999
R4. Design Patterns, Gamma et. al., Pearson Education, 2006.

Lecture Plan:

Lecture#	Topics to be covered	Reference	Learning Objective(s)
01 - 03	Introduction, Object Oriented Information systems Overview- Object Orientation	T1.Ch01 + Class Notes	Basic Concepts of Object Orientation
03 - 04	Software Development Life Cycle Object Oriented Systems Development	T1.Ch 21 + Class Notes	Stages in Software Development & Models
05 - 07	Object Oriented Methodologies Unified Approach	T1.Ch 21, 22	Why unified approach is required?
08 - 09	Unified Modeling Language UML Notations	Class Notes	How to draw & interpret UML diagrams?





Lecture#	Topics to be covered	Reference	Learning Objective(s)
10 - 15	Object Oriented Analysis, Use Case Model, Class Identification, CRC cards, Domain Model, Activity Diagrams, System Design	T1.Ch 2,10,11,12,13, 14 and Case Studies	Systems Analysis using Object Oriented approach
16 - 21	Object Oriented Design, Class Design, Advanced Class Modeling, Design Axioms and GRASP Patterns, Class Diagram Interaction Diagrams, Advanced Interaction Modeling	T1.Ch 15,3,7, 8 and Case Studies	Systems Design using Object Oriented approach and corresponding UML diagrams
22 - 23	State Modeling, UML State Charts, Advanced State Modeling	T1.Ch 5,6	Use of UML diagrams for system dynamics
24 - 25	Access Layer and View Layer, Object Persistence, UI Design	T1.Ch 19	Use of UML diagrams for system layers and UI
26 - 26	Implementation: Implementation Modeling	T1.Ch 17	How to implement?
27 - 32	Design Patterns and their applications	T1.Ch 20, R4	How to simplify design process using patterns?
33 - 35	Architecture and Framework, Object Oriented Systems Architectures Frameworks	Class Notes	System architecture process
36 - 40	Software Quality Assurance Object Oriented Testing	Class Notes	Testing and assuring quality

Evaluation Scheme :

S#	Evaluation Component	Duration	Weight	Date & Time	Nature
1.	Mid-Semester Test	90 Minutes	25%	Saturday, Oct 10, 2026 FN2 (11:30 AM – 12:30 PM)	Closed Book
2.	Case Studies ^{\$}	During class hour	10%	Before Mid-Semester	Open Book
3.	Project*	Three weeks	30%	After Mid-Semester Test	Open Book
4.	Comprehensive Examination	180 Minutes	35%	Tuesday, Dec 15, 2026 AN (02 PM to 05 PM)	Closed Book

\$ During some classes the case-study will be discussed and students will be asked to answer questions in a sheet. Three case studies will be discussed.

* Details of project including problem specifications, deliverables, team size (if any), and deadlines will be posted (immediately after mid-semester test) on the course website.

Chamber Consultation Hour (Wednesday, 9th Hour, 04.00 PM – 04:50 PM)

Notices will be displayed only on the course webpage hosted on Nalanda LMS or will be communicated by email wherever possible and/or necessary.

Make-up Policy: Make-up will be granted in case of extreme exigencies only.

Note (if any): If the marks-histogram emerges as not skewed, then a student getting marks, say marks m out of maximum marks, MM , such that $(MM*m/Acut) < 15*(MM/100)$, where $Acut$ is the cutoff (recommended as per the histogram) for A grade, may be recommended to be reported an NC.

